

Introduction to Geometry

几何学概论：本课程将涵盖标准高中几何学课程的许多主题，适合 9–12 年级的学生。学生将学习线条，线段，角度，弧形，弦，周长，面积，表面积和体积的概念，以及三角形，圆形，四边形和多边形的几何属性。本课程还将介绍逻辑和数学证明的概念。任课教师将调整课程的节奏，以配合学生掌握材料的能力。**预备知识：**所有学生都应该已学完初级代数。若学生已学完代数，对学习几何这门课有帮助，但并非必须的。掌握 MathCounts 的许多知识对于与学习几何学也大有裨益。**教科书：**本课程使用 Richard Rusczyk 的《解题的艺术：几何学概论》。这本书可以从亚马逊网站上购买，价格为 55 美元。所有学生都应该购买这本书以及圆规和量角器。

Introduction to Geometry: This course will cover many of the topics that are part of the standard curriculum for high school geometry – in a way that is suitable for high school students or advanced middle school students. Students will learn about the concepts of lines, line segments, angles, arcs, chords, perimeter, area, surface area and volume along with geometric properties of triangles, circles, quadrilaterals and polygons. They will be introduced to concepts of logic and mathematical proofs. Because this is the first year geometry is being offered, we will adjust the pace of the course to match students' ability to grasp the material. **Prerequisites:** All students should have completed pre-algebra. Completing algebra is not required but will be helpful to students. Students who have completed MathCounts preparation will have been exposed to many of these concepts and should be appropriately prepared as well. **Textbook:** We will be using The Art of Problem Solving: Introduction to Geometry by Richard Rusczyk. This book is available from Amazon for \$55. All students will be expected to purchase this book as well as a compass and protractor.

SCCS Geometry Course

Syllabus 2020/2021 Academic Year

Lesson 1: Lines, Segments, Angles, and Definitions – Covering the basic foundations for the rest of the course, this lesson will review definitions of one dimensional and two dimensional shapes: lines, rays, segments, secants, tangents, diameters, chords, angles of various types and relationships between angles (vertical angles as an example).

Lesson 2: Angles that are part of line systems – This lesson will teach how to solve the value of an unknown angle when the value of other related angles is known. We will cover parallel lines, solving for the value of angles about transversals of parallel lines, parallelograms, the basics of geometric proofs, sum of the angles of a triangle and solving the value of exterior angles.

Lesson 3: Congruent triangles – This lesson will cover the four theorems of congruent triangles and how to use them in geometric proofs. These theorems will also be used to solve the value of unknown angles. Unique properties of isosceles and equilateral triangles will be introduced.

Lesson 4: Area and perimeter – This lesson will cover the concepts of area and perimeter and how to compute those values for basic shapes (triangles, rectangles and squares). We will also cover how to use these basic concepts to determine the area and perimeter of more complex shapes by breaking them down into basic parts. We will derive the formula for computing the area of a triangle.

Lesson 5: Similar triangles – This lesson will cover the concept of similar triangles and the three theorems of triangle similarity. We will prove each of these triangle similarity relationships. We will also use similar triangle principles to solve the length and angles of triangles that are similar to each other.

Exam 1: To cover all material taught through Lesson 5. (October 13 or October 20)

Lesson 6: The Pythagorean Theorem – This lesson will cover proof of the Pythagorean Theorem, its application to solving the length of triangle sides and its use in other geometric proofs. We will also examine isosceles right triangles, 30/60/90 triangles and Pythagorean triples to rapidly solve problems involving right triangles.

Lesson 7: Right Triangles – This lesson will cover the principle of hypotenuse leg congruence and leg leg congruence for right triangles. Likewise, the extension to hypotenuse leg and leg leg similarity will be covered. We will also cover Heron's formula for computing the area of any triangle using only the length of its sides.

Lesson 8: Special parts of triangles – This lesson will cover perpendicular bisectors, angle bisectors, altitudes and medians and their intersection to form the circumcircle, incircle, centroid and orthocenter

of the triangle. Unique properties of these intersections and shapes and their application to computing geometric problems with triangles will be covered.

Midterm Exam – Covering all material learned thus far with an emphasis on material after Exam 1.

Lesson 9: Quadrilaterals – This lesson will cover the sum of angles for a quadrilateral, trapezoids, computing the area of a trapezoid, special properties of isosceles trapezoids, parallelograms, rhombi, rectangles and squares.

Lesson 10: Polygons – This lesson will cover polygons beyond four sides, to include definition of pentagons through dodecagons, the concept of regular polygons, computing the sum of angles, interior angles and exterior angles of polygons and methods for determining the area of a polygon.

Lesson 11: Geometric inequalities – This lesson will cover triangle inequalities for acute and obtuse triangles and how that can be used to determine the legitimacy of a triangle via the triangle inequality. We will use Pythagorean inequality relationship to determine if a triangle is acute or obtuse. These principles will be used in geometric proofs.

Lesson 12: Circles – This lesson will cover basic principles of circles: computing area and circumference, relationships between circles and triangles, partial areas of shapes inscribed in circles and computing areas of intersecting regions for two circles.

Lesson 13: Inscribed angles – This lesson will cover inscribed angles, angles formed from intersecting lines that extend through a circle, angles formed with tangents, and angles formed between tangents and secants. We will discuss how to use these principles to compute the length of arcs or the measure of inscribed angles as well as their application to geometric proofs.

Lesson 14: Power of a point – This lesson will cover the power of a point theorem and its application to measuring the length of chords/secants that intersect at a point within or outside a circle.

Exam 3 - – Covering all material learned thus far with an emphasis on material after the midterm exam.

Lesson 15: Planes – This lesson will cover the concept of a geometric plane, what happens when planes intersect, defining the angle between intersecting planes. We will also cover 3D shapes, such as prisms, and pyramids, and how to calculate their volumes and surface areas.

Lesson 16: Cylinders – This lesson will cover 3D shapes involving circles: cylinders, cones and spheres. We will cover formulas for computing their surface areas and volumes, which will be derived for cylinders and cones. We will describe the interaction of planes with these 3D shapes.

Lesson 17: Translations, Rotations, Dilations, Reflections – This lesson will introduce the Cartesian coordinate system and how it is used to describe the transposition of 2D and 3D shapes. We will explore how lines of symmetry and these geometric transpositions can be used to simplify geometric problems. We will also prove geometric similarity for dilated shapes.

Lesson 18: Analytic geometry -This lesson will cover the Cartesian coordinate system and how it can be used to simplify geometric proofs and computations. Covering the intersection of geometry and algebra, we will introduce linear equations of different forms, compute the distance between lines, find the midpoint of a line, understand relationships between perpendicular lines. We will also explore equations for circles and learn the concept of completing the square to derive the center and the radius of any circle.

Lesson 19: Introduction to Trigonometry – We will cover the concept of sines, cosines and tangents and use the Law of Sines and the Law of Cosines to solve geometric problems associated with triangles.

Final Exam: Covering all material in the course.